SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	SPARK SURGE
Project / Solution Name	LARIS: Quotation Tracker for Malaysian SMEs
Name (Leader)	HARRISH GOVIND A/L SETHULINGAM
Email & Phone (Leader)	harrishgovind09@gmail.com · [0125220471]
University Affiliation(s)	UNIVERSITI TUN HUSSEIN ONN MALAYSIA
Live Prototype URL	https://laris-malaysia.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/lkaixian/laris

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

LARIS runs as one FastAPI process on a Replit deployment. It serves the React web app, the REST API (/api), the public customer pages (accept quote, sign delivery, track) and the Telegram and WhatsApp webhooks. Supplier files arrive by upload or chat. Text and tables are read directly (pypdf, openpyxl). Scanned images go through Tesseract OCR. Free text and voice notes go to the ILMU API, which returns items and quantities as JSON. Catalog photos go to Qwen2.5-VL-7B on a RunPod GPU running vLLM, and OCR must find the same digits before a photo price is kept. Every model result passes a confidence gate and arithmetic, date and catalogue checks, and uncertain lines are flagged for the owner.

Confirmed lines are written to PostgreSQL. A rules engine in Python prices quotes from the cost book, and no model sets a price (price_src has no AI value). The database enforces row-level security between businesses, owner-only approval, and an append-only audit log, payments and document transitions. APScheduler runs follow-ups, expiry and data retention. Documents go out as PDF, XLSX or CSV, by chat or by email (Mailtrap). The licence is verified with Ed25519 and bought through Stripe Checkout. Keys live only in Replit Secrets.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.13	Backend language	PSF License
FastAPI 0.159.1 (Starlette 1.9.0)	API server, webhooks, serves the web app	MIT / BSD-3-Clause
uvicorn 0.34.0	ASGI web server	BSD-3-Clause

Name & Version	Purpose in Project	License
psycopg 3.3.6	PostgreSQL driver	LGPL-3.0
httpx 0.28.1	HTTP client for ILMU, RunPod, Telegram, WhatsApp, Mailtrap	BSD-3-Clause
APScheduler 3.11.3	Follow-ups, quote expiry, chat retention	MIT
pypdf 6.21.4	Text layer from supplier PDFs	BSD-3-Clause
Pillow 13.0.0	Image decoding and checks	MIT-CMU
pytesseract 0.3.13 + Tesseract OCR	OCR for scanned quotes	Apache-2.0
openpyxl 3.1.5, defusedxml 0.7.1	Read XLSX quotes, BOM export, safe XML parsing	MIT / PSF
reportlab 5.0.1	Customer quote PDF	BSD
cryptography 50.0.2	Ed25519 licence signature check	Apache-2.0 or BSD-3-Clause
PyYAML 6.0.2	Trade templates	MIT
python-multipart 0.0.32	File uploads	Apache-2.0
TypeScript 7.0.2	Frontend language	Apache-2.0
React 19.3.1	Web user interface	MIT
Vite 8.3.5, @vitejs/plugin-react 6.1.2	Frontend build	MIT
Tailwind CSS 4.3.3	Styling	MIT
PostgreSQL 16	Database	PostgreSQL License

Name & Version	Purpose in Project	License
pytest 9.0.3, PGlite 0.5.11, axe-core 4.14.0	Automated tests, in-memory Postgres for tests, accessibility checks (development only)	MIT / Apache-2.0 / MPL-2.0
uharfbuzz 0.57.0	Tamil text shaping in PDFs	Apache-2.0
Noto Sans SC, Noto Sans Tamil (new, Regular, subset)	Chinese and Tamil fonts in PDFs	SIL OFL 1.1
Ubuntu font	Latin and Malay text in PDFs and web	Ubuntu Font Licence 1.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
ILMU API (YTL AI Labs)	ilmu-mini-v3.3 extracts items; ilmu-asr-v4.2 transcribes voice notes	Paid, pay per use
RunPod API	Qwen2.5-VL-7B catalog photos; Malaysian Whisper fallback (code written, endpoint not deployed)	Sponsor-provided credits
Telegram Bot API	Live customer and staff channel	Free
Mailtrap Email API	Sign-up codes, resets, invitations, quote emails	Free tier
Stripe Checkout (Stripe)	One-time licence purchase; signed webhook issues the key	Free (Sandbox Mode) no real payments taken in the demo
NTP servers (pool.ntp.org, time.google.com)	Trusted time for licence check	Free
WhatsApp Business Cloud API, Graph v21.0 (Meta)	Built, not connected to Meta	Free tier; paid per template

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit deployment	Hosts the FastAPI process, web app, webhooks, scheduler	Region Not Stated
Replit PostgreSQL	Main database	Included

Service & Provider	Purpose in Project	Plan / Region
Replit Secrets	All API keys and the database URL	Included
RunPod GPU pod, vLLM (Qwen2.5-VL-7B)	Catalog photos; stopped when idle, about US$0.59 an hour	Pay per hour; International
GitHub and GitHub Actions	Repo, CI (tests, pip-audit)	Free

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
ilmu-mini-v3.3	YTL AI Labs	Pre-trained	Commercial API terms
ilmu-asr-v4.2	YTL AI Labs	Pre-trained	Commercial API terms
Qwen2.5-VL-7B-Instruct	Alibaba Qwen (Hugging Face), on RunPod	Pre-trained	Apache-2.0
Mesolitica Malaysian Whisper	Mesolitica (Hugging Face), RunPod	Pre-trained, fallback only.	Not stated on model card; base Whisper MIT
Tesseract 5 LSTM (English)	tesseract-ocr	Pre-trained	Apache-2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
LARIS synthetic demo seed	https://github.com/lkaixian/laris (db/seed_made_to_order.sql)	Synthetic	Apache-2.0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic supplier quotes and enquiries (8 documents, 3 enquiries)	https://github.com/lkaixian/laris (tests/fixtures/docs)	Synthetic	Apache-2.0	N
Trade templates	https://github.com/lkaixian/laris (config/templates)	Self-made	Apache-2.0	N
LARIS Demo Video Test Case	https://github.com/lkaixian/laris/demo_material	Synthetic	Apache-2.0	N
60 invented enquiries for extraction evaluation	https://github.com/lkaixian/laris/blob/master/tests/fixtures/enquiries_eval.jsonl	Synthetic	Apache-2.0	N
3D-printer parts and consumables seed	https://github.com/lkaixian/laris (db/seed_3dprint_parts.csv and .sql)	Synthetic	Apache-2.0	N
Published statistics, citation only (DOSM, UN SDG metadata, The Star)	dosm.gov.my, unstats.un.org, thestar.com.my	Public	Per source	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code with Claude Sonnet 5.5 models (Anthropic)	Code, tests, bug fixes, documentation, research notes, templates, pitch deck, demo video script	Major
Replit Repo AI Chat	Automation of git pull only.	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every AI-written change runs through 1,123 automated tests and GitHub Actions CI before it is kept, with tests written before the code. We click through screens in a browser, re-run the demo flow, and fix the bugs we find. Teammates' pull requests are read before merge, and review has caught real defects, for example whole quantities printed as 1E+2, and an edge-case test run found three licence bugs. Every figure in our documents traces to a cited source with a confidence label, and unmeasured numbers are marked as not measured. The team reviewed the code and documents and takes responsibility for them.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Hosting platform (Reserved VM, PostgreSQL, Secrets)	Hosts the whole prototype: web app, API, webhooks, scheduler and database
RunPod	GPU cloud credits (Serverless)	Runs Qwen2.5-VL-7B for product photos and the Malaysian Whisper speech-to-text fallback

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is essential as the host of the running prototype, though the app is a standard FastAPI and PostgreSQL service that could move to another VM. RunPod is not essential to the core journey: enquiry to rules-priced quote to owner approval runs without it. It adds catalog-photo reading, and any OpenAI-compatible vision endpoint could replace it.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The prototype holds synthetic data only. In real use it stores business users (name, username, email, hashed password), customer and supplier contacts, chat text, voice-note transcripts and documents. Text, audio and photos go to ILMU, RunPod and Meta or Telegram to be read. Stripe receives the purchaser's email and chosen plan at checkout.

- Protection:
  - scrypt password hashing and hashed session tokens;
  - sign-in throttling;
  - row-level security between businesses;
  - staff cannot see costs;
  - contact details masked before text reaches a model;
  - keys only in Replit Secrets;
  - append-only audit log.
- Customer rights: a consent notice on first chat contact, plus /privacy, /mydata and /forget. A breach register tracks the deadline clocks.
- Retention: chat text is blanked after 180 days. Uploaded files are read in memory and only extracted lines are kept.
- These privacy tools have not been reviewed by a lawyer.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

- Models can misread blurred photos, receipts or Manglish speech. Accuracy on real Malaysian quotes and real voice notes has not been measured.
- Mitigations:
  - models never set prices;
  - every line passes a confidence gate and arithmetic, date and catalogue checks;
  - uncertain lines are flagged and the owner approves before anything is sent;
  - AI calls are logged.
- Single-provider dependency (ILMU, RunPod, Meta) is softened by fallbacks: Whisper on RunPod, Telegram in place of WhatsApp, and manual entry.
- WhatsApp is not connected to Meta.
- The GPU pod has a cold start of a few minutes.
- The e-signature is simulated and labelled so.
- The app runs as one process with no failover, and rate limits and cost are not load-tested.
- The live Replit database has not been verified with the row-level-security role. Row-level security is tested in CI.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

☒ Any use of official SDC sponsor resources has been declared in Section 5.

☒ The prototype described here is accessible to judges through the links provided in Section 1.

☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team’s knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

HARRISH

1/11/2026